



42-MAVZU · V BOB. DRF VA DEPLOY

Yakuniy nazorat ishi — IV–V boblar (Django, DRF, Deploy) va yakuniy loyiha

Django, Django REST Framework va deploy bo'yicha nazorat: 25 ta yopiq test va 5 ta ochiq topshiriq (jami 100 ball); yakuniy loyiha talablari, g'oyalar, baholash mezonlari va taqdimot rejasini.

8-sinf · Web Full-stack dasturlash

Bugun: nazorat ishi va yakuniy loyiha

Qism	Mazmuni	Ball
1-qism	25 ta yopiq test × 3 ball	75
2-qism	5 ta ochiq savol / amaliy topshiriq × 5 ball	25
Jami		100

- Keyin: **yakuniy loyiha** talablari, g'oyalar, baholash va taqdimot rejasi

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Takrorlash

IV–V boblar: Django, DRF, Deploy

Takrorlash rejası

Mavzu	Bilishingiz kerak
32. Python + PostgreSQL	venv, psycopg, commit, parametrli so'rovlar, SQL injection
33–34. Django, URL, View	MVT, startapp, <code>INSTALLED_APPS</code> , <code>path</code> , <code>include</code> , converterlar, <code>Http404</code> , <code>redirect</code>
35. Template, static	<code>render</code> , <code>{{ }}</code> , <code>{% %}</code> , filtrlar, <code>extends/block/include</code> , <code>{% url %}</code> , <code>{% static %}</code>
36. Model, ORM	Maydonlar, <code>ForeignKey</code> , <code>makemigrations/migrate</code> , <code>filter/get/create</code> , lookups
37–38. Admin, Forms, CRUD	<code>ModelAdmin</code> , <code>ModelForm</code> , <code>is_valid</code> , <code>csrf_token</code> , <code>login_required</code> , <code>instance=</code> , <code>Paginator</code>
39–40. DRF	REST, status kodlar, Serializer, <code>ViewSet</code> , Router, permissions, CORS
41. Deploy	<code>DEBUG=False</code> , <code>SECRET_KEY</code> , <code>ALLOWED_HOSTS</code> , gunicorn, whitenoise, <code>collectstatic</code>

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Namunaviy testlar

Avval o'ylang — keyin javobni ko'ring

Savol

SQL injection'dan himoyalangan so'rov qaysi?

A `cur.execute(f"... WHERE ism='{ism}')`

B `cur.execute("... WHERE ism=%s",
(ism,))`

C `cur.execute("... WHERE ism='" + ism +
"')`

D Hammasi xavfsiz

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

`path("kitob/<int:pk>/", views.kitob)` — view qanday e'lon qilinadi?

A `def kitob(request):`

B `def kitob(pk):`

C `def kitob(request, id):`

D `def kitob(request, pk):`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

"Kitobning bitta muallifi, muallifning ko'p kitobi" — Kitob modelida qaysi maydon?

A `ManyToManyField(Muallif)`

B `CharField`

C `ForeignKey(Muallif, on_delete=...)`

D `OneToOneField`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Narxi 50 000 dan kichik kitoblar?

- A** `Kitob.objects.get(narx<50000)`
- B** `Kitob.objects.filter(narx__lt=50000)`
- C** `Kitob.objects.all(narx=50000)`
- D** `Kitob.filter("narx < 50000")`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

REST API'da resursni o'chirish metodi va javob kodi?

A GET, 200

B POST, 201

C DELETE, 204

D PUT, 404

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Boshqa domendagi JS sayt API'ga murojaat qilishi uchun Django'da nima sozlanadi?

A CSRF

B ALLOWED_HOSTS

C CORS (django-cors-headers)

D WhiteNoise

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Internetdagi saytda qaysi sozlama **noto'g'ri**?

A `DEBUG = False`

B `DEBUG = True`

C `ALLOWED_HOSTS = ["sayt.onrender.com"]`

D `SECRET_KEY` muhit o'zgaruvchisida

Javobni tanlang — to'g'ri variant yashil bo'ladi

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Amaliy qism

5 ta topshiriq × 5 ball

1-topshiriq: /maqola/5/ ochilganda nima bo'ladi?

- 1 So'rov Django'ga keladi
- 2 **urls.py** `maqola/<int:id>/` ni topadi → view'ni `id=5` bilan chaqiradi
- 3 **View** → **Model** (ORM): `get_object_or_404(Maqola, id=5)`; topilmasa — **404**
- 4 View maqolani **Template**'ga beradi — `render(...)`
- 5 Shablon to'ldiriladi → tayyor **HTML** brauzerga qaytadi

2-topshiriq: model va admin

Python

```
class Kitob(models.Model):
    JANRLAR = [("roman", "Roman"), ("shariyat", "She'riyat"), ("ilmiy", "Ilmiy")]
    nomi = models.CharField(max_length=200)
    muallif = models.CharField(max_length=100)
    janr = models.CharField(max_length=20, choices=JANRLAR)
    yil = models.PositiveIntegerField()
    narx = models.DecimalField(max_digits=10, decimal_places=2)
    mavjud = models.BooleanField(default=True)
    qoshilgan = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return self.nomi

@admin.register(Kitob)
class KitobAdmin(admin.ModelAdmin):
    list_display = ["nomi", "muallif", "janr", "mavjud"]
    list_filter = ["janr", "mavjud"]
    search_fields = ["nomi", "muallif"]
```

3-topshiriq: view va shablon

Python

```
def kitoblar(request):
    q = request.GET.get("q", "").strip()
    qs = Kitob.objects.filter(mavjud=True)
    if q:
        qs = qs.filter(nomi__icontains=q)
    return render(request, "kutubxona/kitoblar.html", {"kitoblar": qs, "q": q})
```

Django shablon

```
{% extends "kutubxona/base.html" %}
{% block content %}
    <form><input name="q" value="{{ q }}"><button>Izlash</button></form>
    <ul>
        {% for k in kitoblar %}
            <li><a href="{% url 'kutubxona:kitob' k.pk %}">{{ k.nomi }}</a> – {{ k.muallif }}</li>
        {% empty %}
            <li>Kitob topilmadi</li>
        {% endfor %}
    </ul>
{% endblock %}
```

4–5-topshiriqlar: API va deploy

4: DRF API

```
class KitobSerializer(serializers.ModelSerializer):
    class Meta:
        model = Kitob
        fields = "__all__"
    def validate_narx(self, v):
        if v <= 0:
            raise serializers.ValidationError("Narx musbat bo'lsin")
        return v

class KitobViewSet(viewsets.ModelViewSet):
    queryset = Kitob.objects.all()
    serializer_class = KitobSerializer
    permission_classes = [permissions.IsAuthenticatedOrReadOnly]
    filter_backends = [filters.SearchFilter]
    search_fields = ["nomi"]

router = DefaultRouter()
router.register("kitoblar", KitobViewSet)
```

• 5: Deploy bosqichlari

- 1 `gunicorn`, `whitenoise`, `django-database-url` + `requirements.txt`
- 2 `DEBUG=False`, `SECRET_KEY` va baza — muhitdan; `ALLOWED_HOSTS`, `CSRF_TRUSTED_ORIGINS`
- 3 `build.sh`: `install` → `collectstatic` → `migrate`; `push`
- 4 Render: Postgres + Web Service, `gunicorn` `config.wsgi:application`
- 5 Superuser, tekshiruv ro'yxati, Logs

Baholash shkalasi

Ball	Natija
86–100	A'lo — kurs dasturini to'liq o'zlashtirgansiz
71–85	Yaxshi
60–70	Qoniqarli
0–59	Qoniqarsiz — mentor bilan takrorlash

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Yakuniy loyiha

31-loyiha: kurs davomida o'rganilgan hamma narsa bitta loyihada

Majburiy talablar

Qism	Talab
Front-end	Semantik HTML, o'z CSS dizayni, responsiv, JS interaktivlik
Ma'lumotlar bazasi	PostgreSQL; 3+ model, kamida bitta <code>ForeignKey</code>
Back-end	Django CRUD (5 sahifa), shablon merosi, formalar, admin
Foydalanuvchilar	Ro'yxatdan o'tish / kirish; "faqat muallif" ruxsati
API	Kamida bitta resurs uchun DRF API (GET + POST)
Algoritmik qism	Statistika, reyting, hisob-kitob, filtrlash
Git va deploy	15+ mazmunli commit, README, internetga joylangan

- Individual yoki 2–3 kishilik jamoada

Loyiha g'oyalari

-  **Maktab kutubxonasi:** katalog, ijara, jarima, eng ko'p o'qilganlar reytingi
-  **Onlayn test tizimi:** o'qituvchi test yaratadi, o'quvchilar yechadi, reyting
-  **Sinf jurnali:** o'quvchilar, fanlar, baholar, davomat, hisobotlar
-  **Maktab do'koni / yarmarka:** mahsulotlar, savat, buyurtmalar, ombor
-  **To'garaklar portali:** joylar soni, ro'yxatdan o'tish, jadval
-  **O'z g'oyangiz** — mentor bilan kelishib

Bajarish rejasi (4 hafta)

- 1 **1-hafta — reja:** funksiyalar, ER-diagramma, Figma maket, GitHub + Kanban
- 2 **2-hafta — back-end:** modellar, migratsiyalar, admin, CRUD, formalar
- 3 **3-hafta — front-end:** shablonlar, dizayn, responsivlik, JS
- 4 **4-hafta — API, sinov, deploy:** DRF, testlar, Render, README
- 5 **Taqdimot**

Baholash mezonlari (100 ball)

Mezon	Ball
Funksionallik — barcha talablar xatosiz ishlaydi	30
Kod sifati — nomlar, tuzilma, xavfsizlik (CSRF, ruxsatlar)	20
Dizayn va qulaylik — chiroyli, responsiv	15
Ma'lumotlar bazasi tuzilmasi	10
Git, README va deploy	10
Taqdimot — tushuntirish va savollarga javob	15

Taqdimot rejasi (5–7 daqiqa)

- 1 **Muammo va g'oya** — kimga va nima uchun kerak? (30 soniya)
- 2 **Jonli namoyish** — internetdagi sayt, asosiy ssenariylar (3 daqiqa)
- 3 **Texnik tuzilma** — modellar diagrammasi, texnologiyalar (1–2 daqiqa)
- 4 **Qiyinchiliklar va xulosalar** — eng qiyin muammo va yechim (1 daqiqa)
- 5 **Savollarga javob**
 - Topshirish: GitHub repozitoriyasi yoki mentor tasdiqlagan platforma

Tabriklaymiz — kurs yakunlandi!

**HTML → CSS → JavaScript → Python → PostgreSQL →
Django → DRF → Deploy**

- 66 soatda birinchi tegdan internetda ishlaydigan **full-stack ilovagacha**
- **Keyingi qadamlar:**
 - "Full-stack qo'llanma" kursi: Node.js, Express, React
 - Telegram botlar (Python + Aiogram)
 - Yangi loyihalar — portfoliongizga yangi qadam
 - Olimpiadalar, xakatonlar, IT-tanlovlar



Savollar bormi?

Kurs yakunlandi — tabriklaymiz! 🎉

Batafsil matn, misollar va testlar: **42-nazorat-ishi-3.html** darsida